

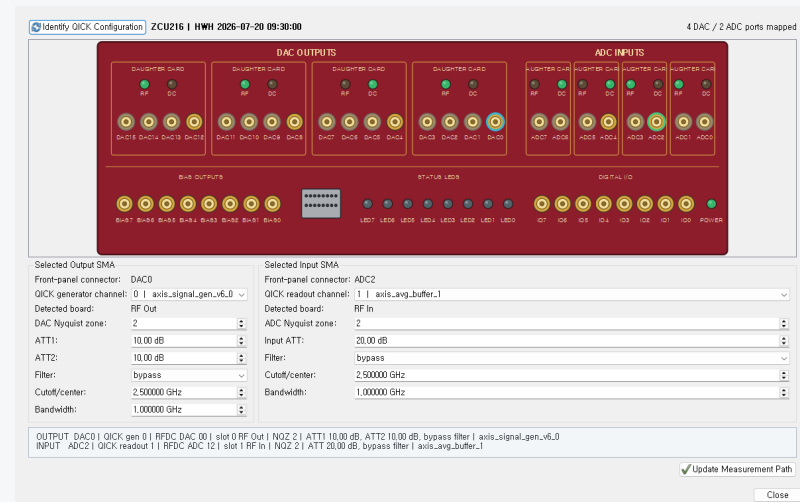
PulseGenerator GUI

User Guide for QICK-Based Pulse and RF Measurements

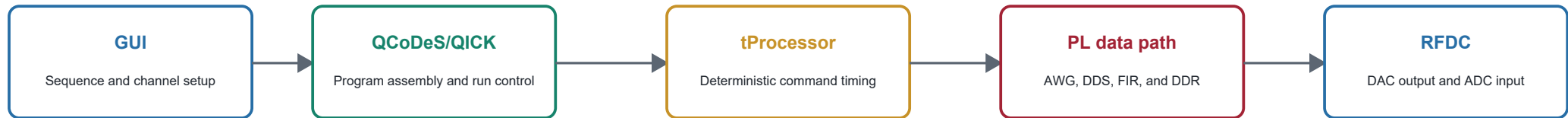
AWG tuning, RF output/readout, FIR-DDR capture, stability diagrams, S-parameters, calibration, and QCoDeS storage

Audience: experimentalists familiar with RF measurements who are new to QICK

Version 1.0 | 20 July 2026



1. What This Application Controls



Control travels left to right; captured FIR I/Q and metadata return to the host.

QICK in one paragraph

QICK combines a software-defined experiment description with FPGA data paths. The tProcessor dispatches timestamped commands. Generator IP drives RFDC DAC channels, readout IP processes RFDC ADC channels, and custom FIR/DDR logic stores long traces. The GUI builds this configuration and keeps channel routing, RF-board settings, pulse timing, and database metadata together.

- The GUI does not stream every waveform point from the PC in real time.
- Pulse commands and sweep parameters are compiled before execution.
- FIR I/Q data returns after FPGA processing and DDR capture.

Major hardware terms

- **RFDC**: integrated DAC and ADC tiles on the ZCU216.
- **tProcessor**: deterministic timing engine that issues generator and trigger commands.
- **Fabric clock**: PL clock used by AXIS generators and custom IP.
- **HWH**: hardware metadata describing instantiated IP and signal connectivity.
- **FIR-DDR**: filtered, decimated capture path that writes I/Q samples into PL DDR.

Firmware assumption

This guide targets the qstl_awg_tuning_fir image and its custom AWG tuning and FIR-DDR path. A different bitstream can expose a different channel map or feature set.

2. Installation and First Connection

Recommended host environment

Use the repository requirements in a dedicated Conda environment. The GUI needs PyQt5, pyqtgraph, NumPy, QCoDeS, Pyro, and the matching local QICK package.

Typical launch

```
conda activate qick_test
python DCWaveformGeneratorGUI/DCWaveform_Generator.py
```

QICK server

Start the Pyro nameserver and QICK server on the ZCU216, then enter the host, nameserver port, and proxy name in AWG Tuning > Experiment.

Before energizing a device

1. **Load firmware** - Confirm the intended BIT/HWH pair is active.
2. **Identify configuration** - Open the Front Panel and read live HWH routing.
3. **Verify daughter cards** - Confirm RF_Out, DC_Out, RF_In, or DC_In for each connector.
4. **Set RF protection** - Start with high attenuation and a conservative DAC gain.
5. **Dry run** - Use Show Program before the first hardware execution.

Clock defaults

The GUI defaults the AWG fabric clock and tProcessor time base to 300 MHz. Keep these values consistent with the deployed firmware and measured timing.

tProcessor time



Instruction timestamp

AWG fabric



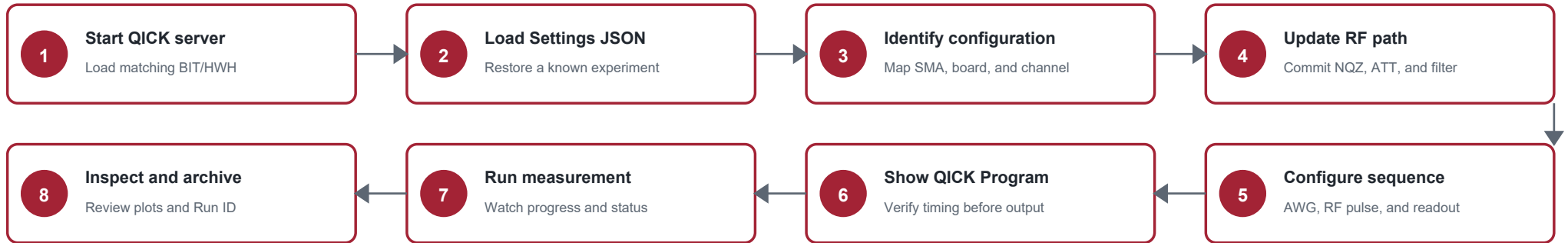
One AXIS word con

FIR DDR output



Stored post-FIR sam

3. Quick Start: Setup Order



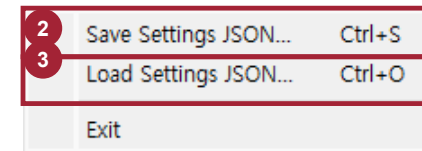
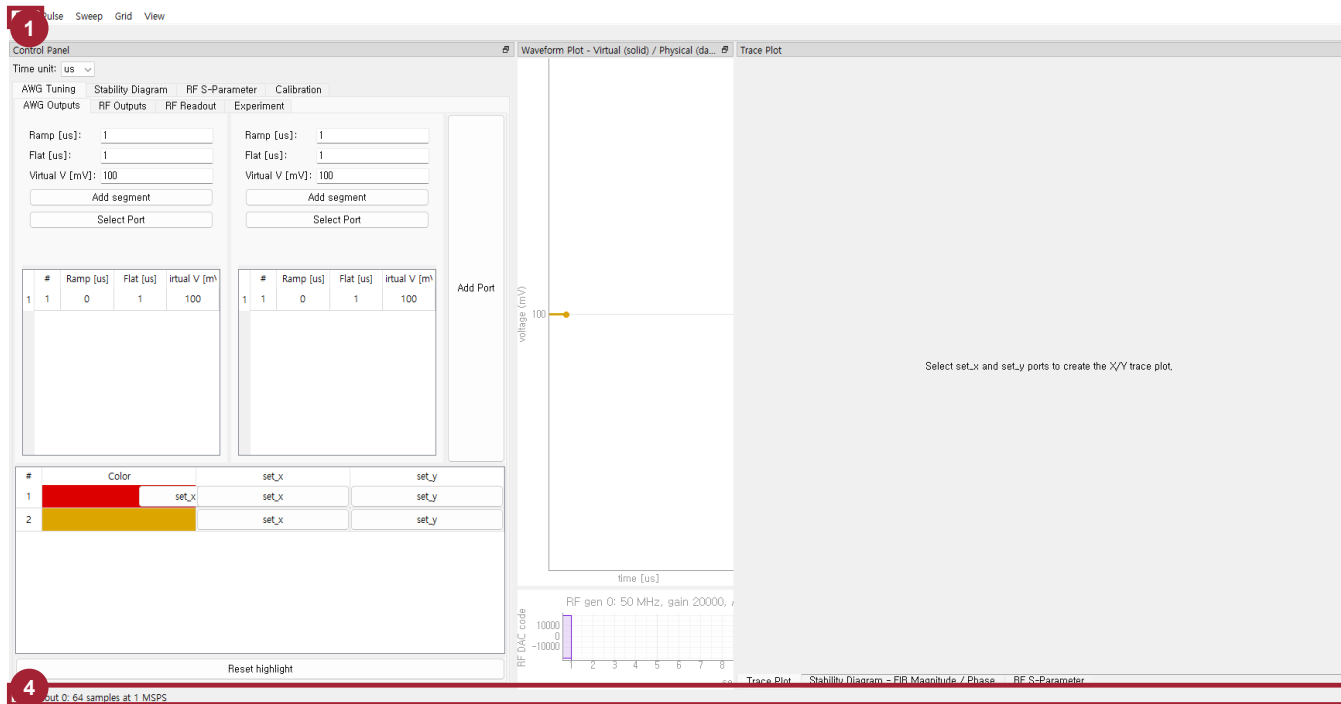
- 1 Start the hardware service**
Start the Pyro nameserver and QICK server with the intended BIT/HWH pair.
- 2 Load a known JSON**
Use File > Load Settings JSON to restore waveform, RF, database, and display settings.
- 3 Identify QICK Configuration**
Read live HWH routing and detected daughter-card types.
- 4 Update Measurement Path**
Commit SMA mapping, channel, board, NQZ, attenuation, filter, and losses.

- 5 Configure the sequence**
Build AWG segments, RF output pulses, and the FIR-DDR readout window.
- 6 Show QICK Program**
Inspect generated instructions and timing before energizing the DUT.
- 7 Run**
Start the requested experiment and watch progress/status until completion.
- 8 Inspect and save**
Review plots, QCoDeS Run ID, metadata, and the status line before changing hardware.

Recommended practice

Complete steps 1-6 with conservative gain and high attenuation. Save a new JSON only after the channel map and timing have been verified.

4. Main Window, Save, and Load



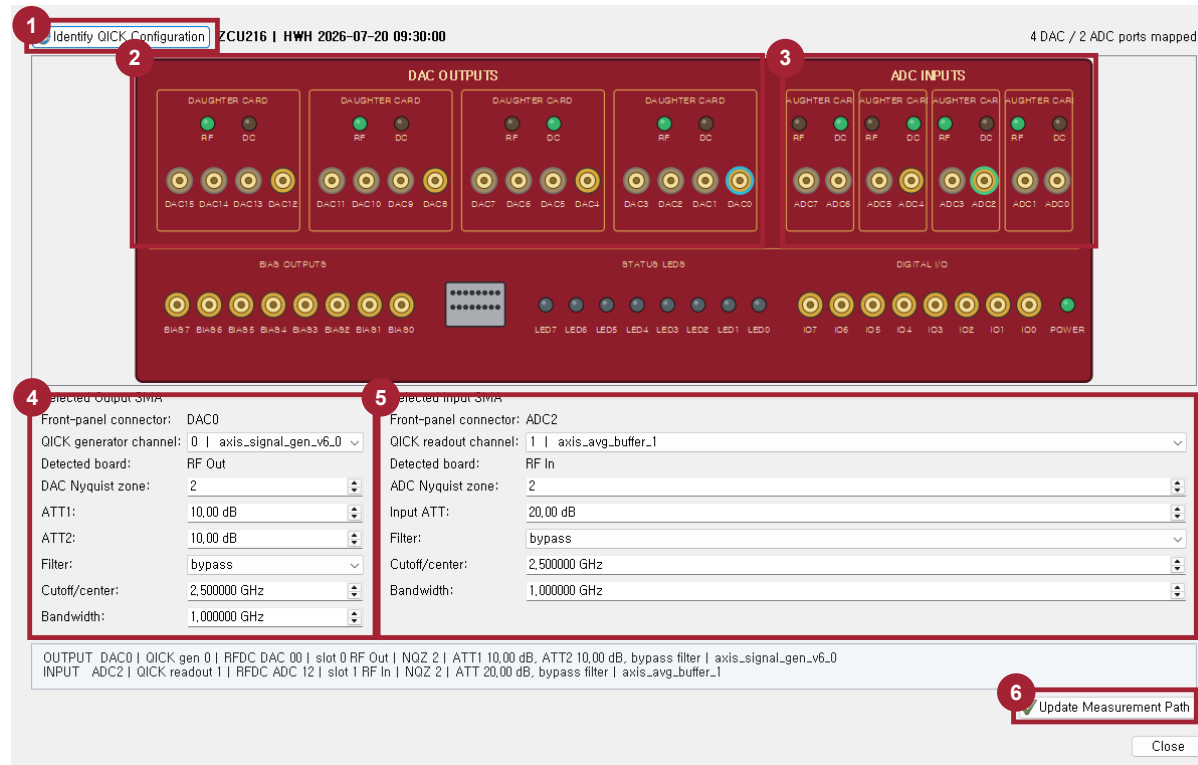
- 1 Open File**
Click File in the top-left menu bar.
- 2 Save**
Click Save Settings JSON... or press Ctrl+S.
- 3 Load**
Click Load Settings JSON... or press Ctrl+O.
- 4 Confirm**
Read the bottom status bar for the saved or loaded path.

JSON contents

A settings file stores AWG vertices, sweeps, cross-capacitance, RF output/readout, shared RF path, calibration and database paths, clocks, compensation, and display preferences. Loading an older schema fills new fields with safe defaults; saving again upgrades it.

5. HWH-Backed Front Panel

Follow the numbered controls from live HWH discovery to a committed RF path.



1 Identify

Click Identify QICK Configuration to read HWH and daughter-card IDs.

2 Select output SMA

Click the physical DAC connector that drives the DUT.

3 Select input SMA

Click the physical ADC connector returning from the DUT.

4 Configure output

Confirm generator, board, NQZ, ATT1/ATT2, and filter.

5 Configure input

Confirm readout, board, NQZ, input ATT or DC gain, and filter.

6 Commit path

Click Update Measurement Path to propagate settings to all measurement tabs.

Nyquist zones

NQZ changes RFDC reconstruction or sampling-zone configuration, not the logical sweep frequency. Confirm the analog band and anti-alias filtering before selecting zone 2.

6. AWG Waveform Authoring

The screenshot shows the 'AWG Tuning' tab in the PulseGenerator GUI. The interface is divided into several sections. At the top, there are tabs for 'AWG Tuning', 'Stability Diagram', 'RF S-Parameter', and 'Calibration'. Below these are sub-tabs for 'AWG Outputs', 'RF Outputs', 'RF Readout', and 'Experiment'. The main area contains two columns of input fields for 'Ramp [us]', 'Flat [us]', and 'Virtual V [mV]'. Below these are 'Add segment' and 'Select Port' buttons. A large table with columns '#', 'Ramp [us]', 'Flat [us]', and 'Virtual V [mV]' is in the center. To the right of this table is an 'Add Port' button. At the bottom, there is a 'Color' column and 'set_x'/'set_y' buttons for each row. A 'Reset highlight' button is at the very bottom.

1. Add Port

2. Enter segment values

3. Add segment

4. Edit a row

5. Choose plot axes

- 1 Add Port**
Create up to eight AWG output editors.
- 2 Enter segment values**
Set ramp time, flat time, and virtual target voltage.
- 3 Add segment**
Append the entered SET/RAMP pair to the selected port.
- 4 Edit a row**
Right-click a segment row to insert, delete, or configure a voltage sweep.
- 5 Choose plot axes**
Use `set_x` and `set_y` to select traces for the X/Y plot.

Cross-capacitance

Use Pulse > Virtual-to-physical cross-capacitance matrix... to map virtual electrodes to physical DAC voltages. Review solid virtual and dashed physical traces before execution.

Grid

Use Grid > Configure grid... and optionally enable Snap dragged points.

7. Sweeps and Bias-T Compensation

1

#	Ramp [us]	Flat [us]	Virtual V [mV]
1	0	1	100

Reset highlight

- 1 Right-click a SET row**
Use the boxed segment table, not the waveform line.
- 2 Open Voltage sweep**
Choose Voltage sweep > Configure sweep... from the context menu.
- 3 Enter the range**
Set start voltage, stop voltage, and point count, then accept the dialog.
- 4 Verify the envelope**
The GUI redraws the extreme traces; adjacent ramps follow the swept SET automatically.

Multiple sweeps

Configure additional output/segment rows the same way. The GUI creates a Cartesian product and nests repetitions inside each sweep point.

DC compensation

In AWG Tuning > Experiment, enable Bias-T compensation, choose DC compensation, then select fixed voltage or fixed time. The complementary value is calculated before program generation.

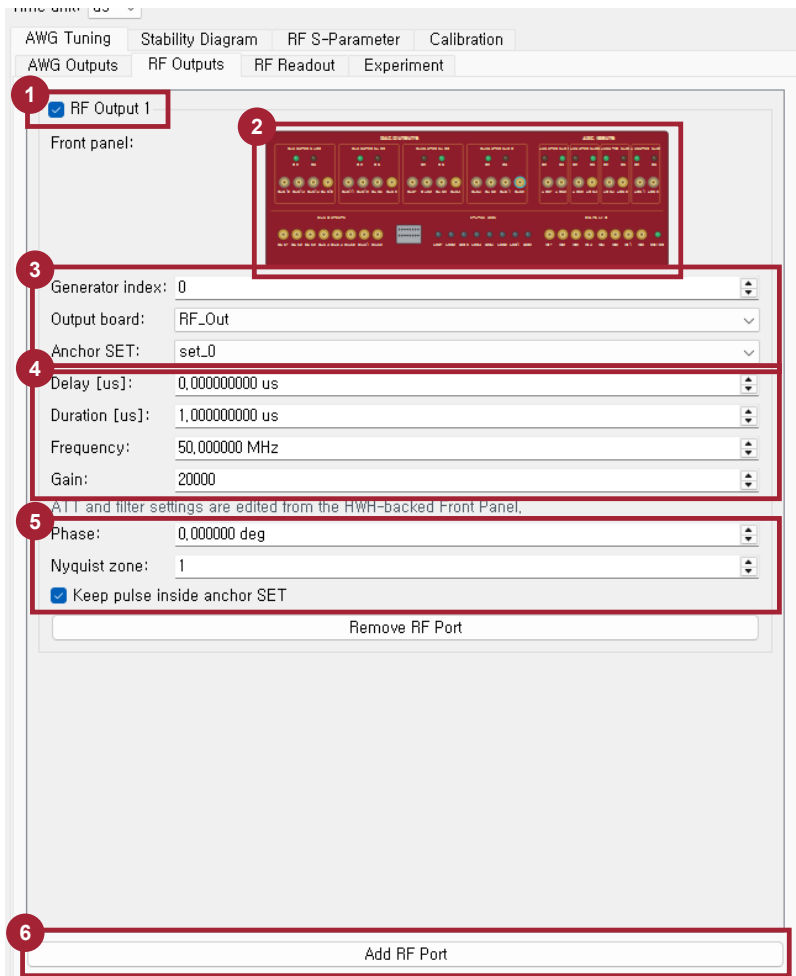
Filter compensation

Choose Filter compensation and enter tau. Flat segments receive target/tau slew-rate correction. Use Show QUICK Program before running.

Safety

Compensation can increase peak voltage or sequence duration. Verify the physical traces and full-scale limits before output.

8. RF Output Channels

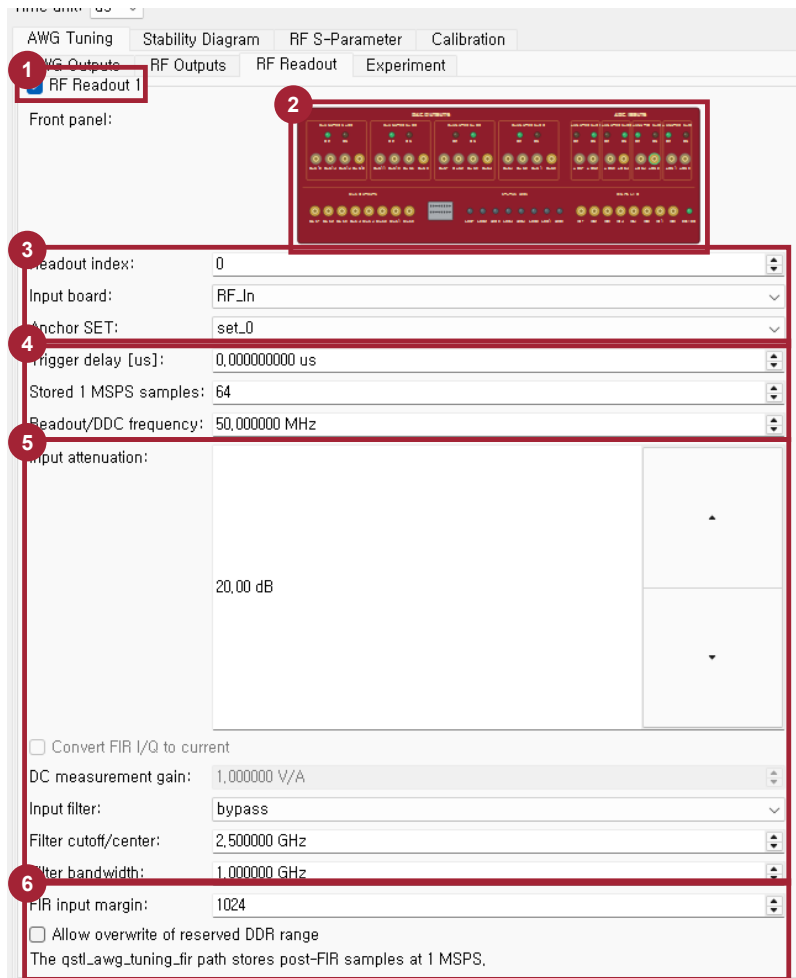


- 1 Enable RF Output**
Check the channel group before editing or running.
- 2 Open the Front Panel**
Click the panel preview to select SMA, board, ATT, filter, and NQZ.
- 3 Select routing**
Set generator index, output board, and anchor SET.
- 4 Set the pulse**
Enter delay, duration, frequency, and gain.
- 5 Finish timing**
Set phase/NQZ and keep the pulse inside its anchor SET when required.
- 6 Add another RF port**
Use Add RF Port for another generator channel.

Gain limit

Start with conservative gain and high attenuation. The software rejects settings above the configured maximum.

9. RF Readout and FIR-DDR Capture



1 Enable RF Readout

Check the readout group.

2 Open the Front Panel

Select ADC SMA, board, NQZ, input ATT or DC gain, and filter.

3 Choose routing

Set readout index, input board, and anchor SET.

4 Define capture

Set trigger delay, stored 1 MSPS sample count, and DDC frequency.

5 Set analog conversion

For DC_In, optionally convert FIR I/Q to current using the configured gain.

6 Set FIR-DDR safety

Choose FIR warm-up margin and enable overwrite only for a safe reserved range.

Stored data

The deployed design filters continuously, aligns decimation phase to the trigger, and stores post-FIR I/Q at nominally 1 MSPS.

10. Running an AWG Tuning Experiment

The screenshot shows the 'AWG Tuning' tab in the PulseGenerator GUI. The configuration is as follows:

Parameter	Value
QICK IP/host:	192.168.2.99
Pyro nameserver port:	8888
Pyro proxy name:	myqick
QCoDeS DB file:	C:\Users\박정현\qick_experiments.db
Experiment name:	QICK pulse experiment
Sample name:	PulseGenerator
AWG fabric clock:	300,000,000 MHz
tProcessor clock:	300,000,000 MHz
AWG full scale (+/-):	2500,000,000 mV
AWG generator indices:	1, 3
Repetitions per sweep point:	1

Below the timing parameters, there is a section for 'Bias-T Compensation' with the following settings:

Parameter	Value
Compensation type:	DC compensation
DC control mode:	Fixed voltage (adjust time)
DC voltage:	250,000,000 mV
DC time:	1,000,000 us
Filter time constant (tau):	100,000,000 us

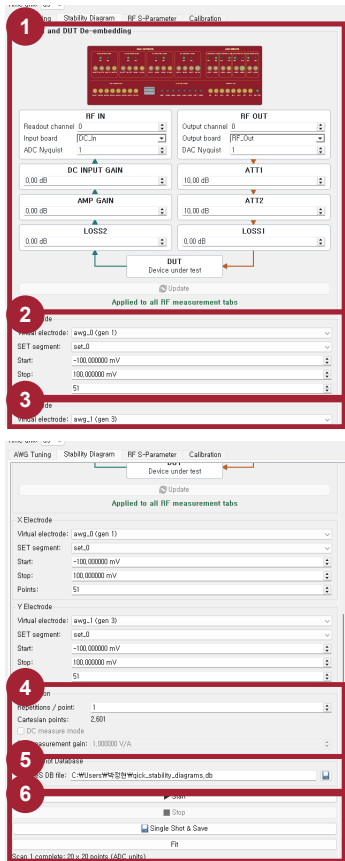
At the bottom, there is a 'Notes' field with the text 'Optional experiment notes'. Below the notes, there are two buttons: 'Run QICK Experiment' (highlighted with a red box and number 5) and 'Show QICK Program' (highlighted with a red box and number 1). The status bar at the bottom shows 'Ready'.

- 1 Connect**
Enter QICK host, nameserver port, and proxy name.
- 2 Choose storage**
Select the QCoDeS DB, experiment name, and sample name.
- 3 Confirm timing**
Set fabric/tProcessor clocks, full scale, AWG indices, and repetitions.
- 4 Show QICK Program**
Inspect assembly and timing before hardware execution.
- 5 Run QICK Experiment**
Start the full sweep/repetition job and watch progress/status.

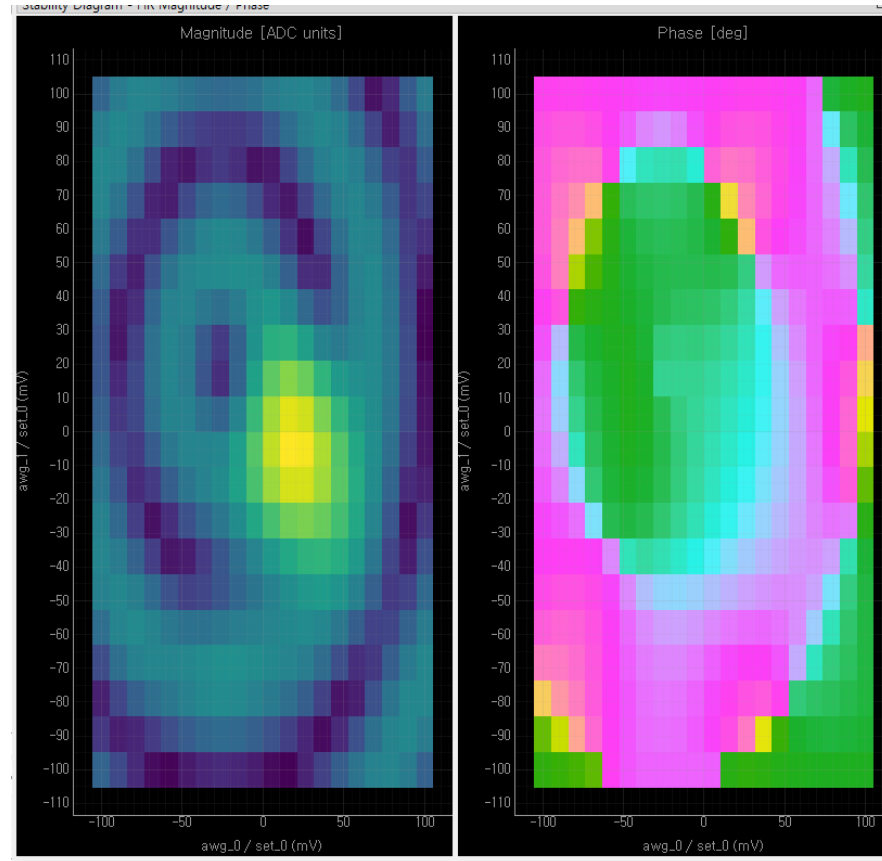
Large data sets

Write to a local SSD, checkpoint SQLite WAL at completion, then copy to synchronized storage. Traces are stored as arrays rather than one SQL row per sample.

11. Stability Diagram

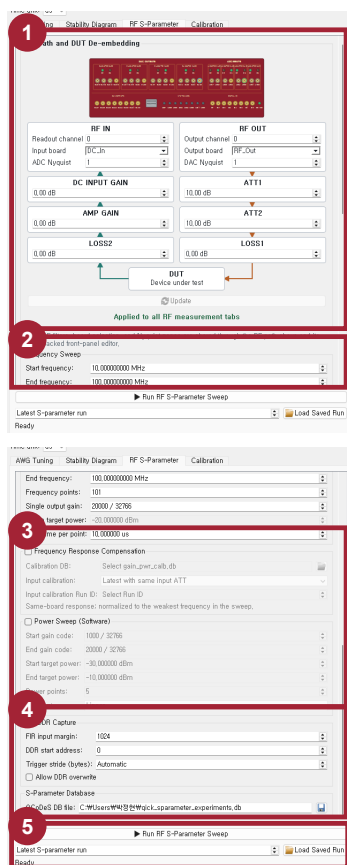


- 1 **Set RF path**
Use the shared output/readout path and commit it first.
- 2 **Set X electrode**
Choose output, SET segment, start/stop voltage, and points.
- 3 **Set Y electrode**
Choose the second output and Cartesian sweep range.

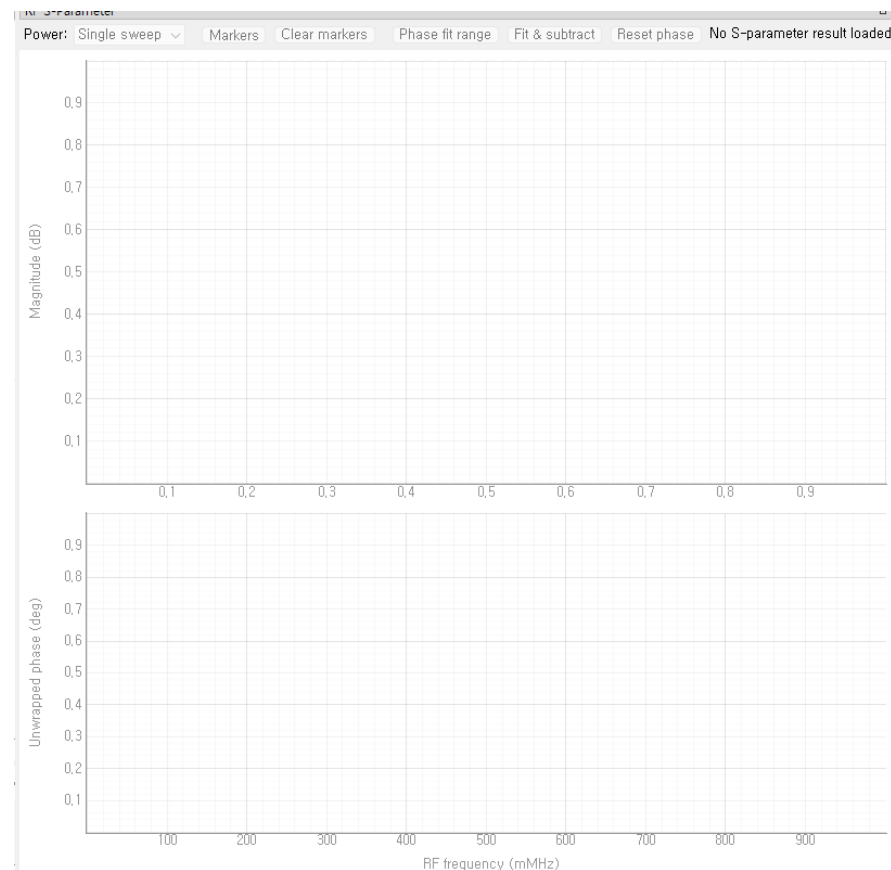


- 4 **Set acquisition**
Choose repetitions per point and optional DC current conversion.
- 5 **Choose single-shot DB**
This database is used only by Single Shot & Save.
- 6 **Start or save**
Start repeats without DB writes; Stop ends after the active scan; Single Shot & Save stores one map.

12. RF S-Parameter Sweep

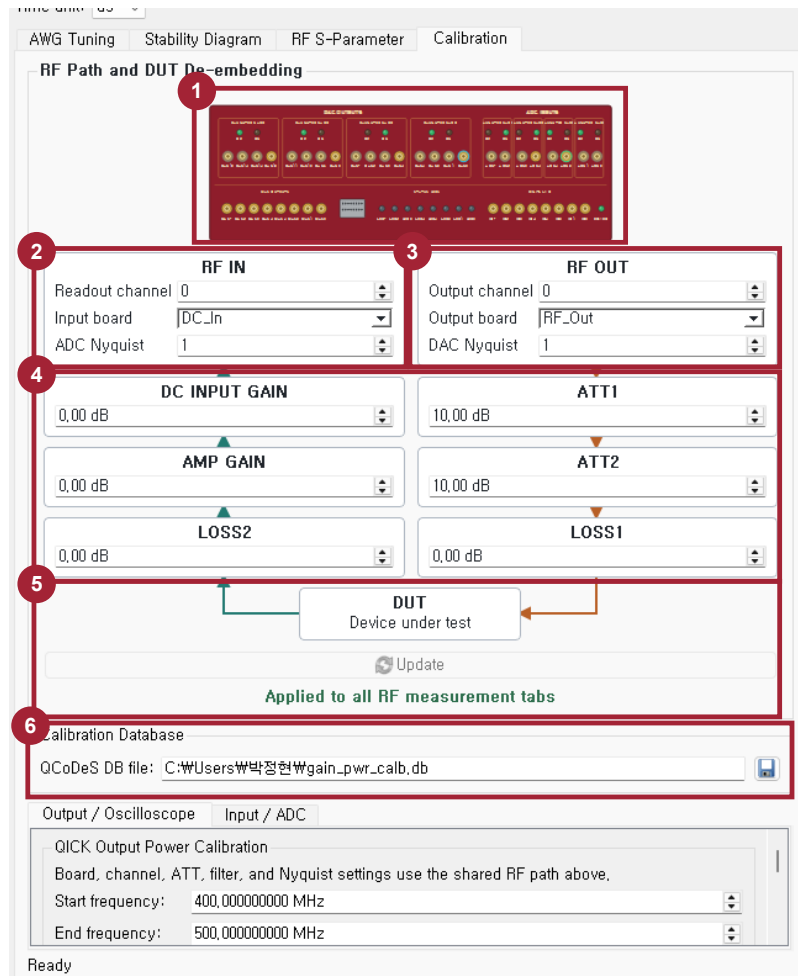


- 1 Set RF path**
Commit output/readout SMA, boards, NQZ, ATT, filters, and de-embedding terms.
- 2 Set frequency sweep**
Enter start/end frequency, point count, gain or target power, and scan time.
- 3 Optional correction**
Enable frequency-response compensation and/or the outer software power sweep.



- 4 Configure capture and DB**
Set FIR margin, DDR address/stride/overwrite, and the independent S-parameter DB.
- 5 Run or reload**
Click Run RF S-Parameter Sweep, or select a Run ID and click Load Saved Run.

13. RF Path Correction and Calibration



- 1 Open the front panel**
Click the QICK image to select HWH-backed SMA and board settings.
- 2 Select RF IN**
Choose the readout channel, input board, and ADC NQZ.
- 3 Select RF OUT**
Choose the output channel, output board, and DAC NQZ.
- 4 Enter path terms**
Set input ATT/amp gain/LOSS2 and output ATT1/ATT2/LOSS1.
- 5 Update**
Commit the path for S-parameter, Stability, Experiment, and Calibration.
- 6 Choose calibration DB**
Use one shared gain/power calibration database.

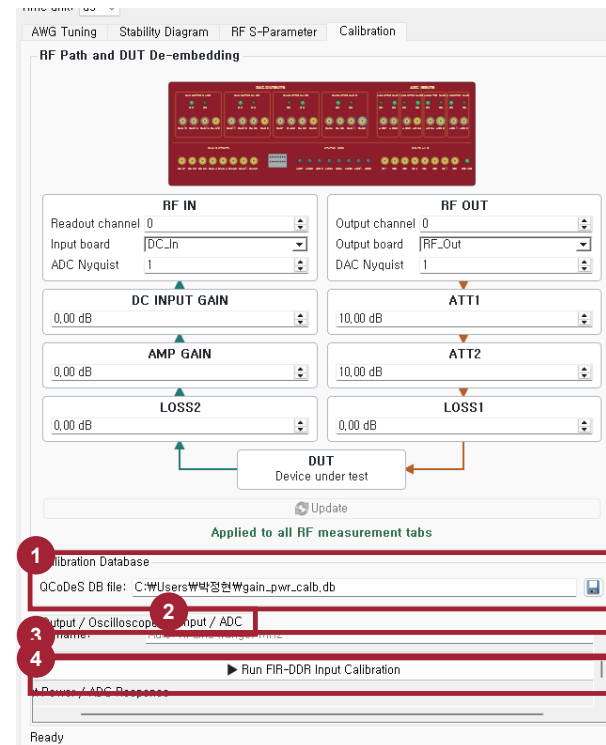
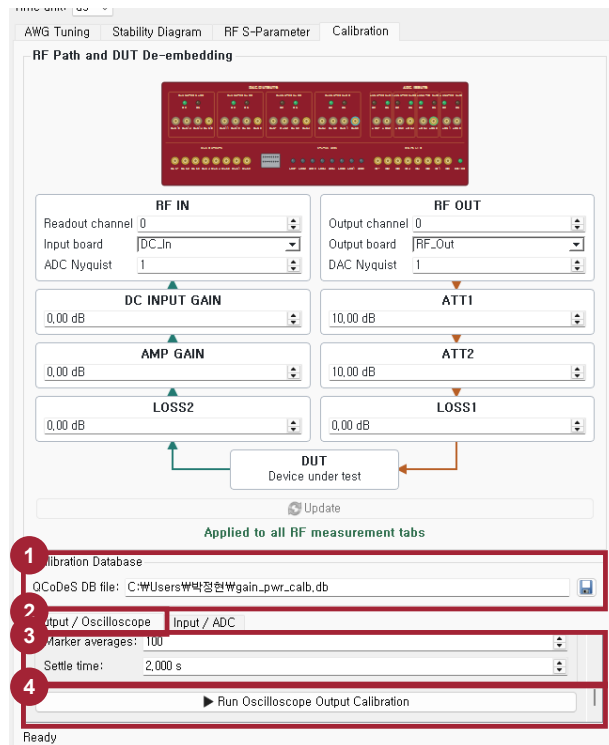
De-embedding equations

- $P_{DUT,in} = P_{RF,OUT} - LOSS1$
- $P_{DUT,out} = P_{RF,IN} + LOSS2 - AMP_GAIN$
- $S21 = P_{DUT,out} - P_{DUT,in}$

Calibration lookup

Board type, frequency overlap, and input attenuation must be compatible. More than 4096 frequency points are compressed to DMEM capacity.

14. Calibration Workflows



- 1 **Choose calibration DB**
Select the QCoDeS gain/power database.
- 2 **Output / Oscilloscope**
Open this tab for absolute output-power calibration.
- 3 **Set scope controls**
Enter frequency/gain axes, VISA resource, FFT marker, averaging, and settle time.
- 4 **Run output calibration**
Click Run Oscilloscope Output Calibration.

- 1 **Use the same DB**
Input and output calibration runs share the selected file.
- 2 **Input / ADC**
Open this tab for input-power versus FIR-DDR ADC response.
- 3 **Set scan metadata**
Confirm frequency/gain scan and experiment/sample names.
- 4 **Run input calibration**
Click Run FIR-DDR Input Calibration; use the plot controls afterward.

Physical S21

With compatible output and input calibrations, the GUI reports calibrated DUT input/output power. Otherwise the result is a relative ADC response, not an absolute VNA-grade S21.

15. QCoDeS Data Layout

Per-shot coordinates

- Sweep coordinates identify output and SET segment, not generic sweep0/sweep1 labels.
- repetition_index identifies the repeated shot inside one Cartesian sweep point.
- I trace and Q trace are stored as full-length one-dimensional arrays.

AWG vertices

- Each AWG channel stores virtual and physical vertex traces separately.
- Vertex time is reconstructed from paired time and voltage arrays; it is not exposed as a sweep selector.

Metadata

- BIT/HWH identity, clocks, channel map, board types, NQZ, attenuation, filters, and RF settings.
- Pulse sequence, cross-capacitance matrix, compensation mode, sweep definition, and repetitions.
- FIR sample rate, sample count, scan timing, calibration Run IDs, and de-embedding terms.

Plottr-inspectr

Select I trace or Q trace as the dependent array and sample index as its array axis. Use sweep coordinates and repetition_index to choose or compare shots.

Database size

Array-per-trace storage avoids millions of scalar SQL rows. Total size still scales with Cartesian points x repetitions x samples x two channels, so estimate storage before a large run.

16. Troubleshooting and Operating Checklist

Common problems

- **Channel appears wrong:** re-run Identify QUICK Configuration and confirm HWH matches BIT.
- **First trace is noisy:** verify arm-before-trigger order, readout warm-up, and trigger offset. Do not assume DDR is the only source.
- **Timing scales differ:** verify the manual tProcessor and fabric clock values against deployed firmware.
- **Attenuator does not respond:** confirm daughter-card seating and detected board type before changing software.
- **No Numpy._core through Pyro:** return plain lists/scalars across RPC or align NumPy serialization versions.

Pre-run checklist

1. **Firmware** - Correct BIT and HWH loaded.
2. **Routing** - Physical SMA, logical channel, board type, and NQZ confirmed.
3. **Protection** - ATT/filter/gain and voltage full-scale reviewed.
4. **Timing** - Pulse, readout, FIR margin, and compensation windows checked.
5. **Storage** - DB path, free space, sweep size, and repetition count checked.
6. **Dry run** - Generated program inspected before energizing the DUT.

Support information

When reporting a problem, include the saved JSON, software commit, BIT/HWH pair, HWH channel summary, QCoDeS Run ID, complete traceback, and the exact settings that produced the issue.